

A Novel Hybrid Quantum Architecture for Path Planning in Quantum-Enabled Autonomous Mobile Robots

Mayukh Sarkar, Jitesh Pradhan[✉], Anil Kumar Singh, *Senior Member, IEEE*, and Hathiram Nenavath[✉]

Abstract—Autonomous mobile robots are being used increasingly as consumer devices around the globe, such as for fetching items and cleaning purposes, to name a few, in households and industries. Such robots are being employed to traverse a set of target locations and provide the necessary services. In a 2-dimensional environment, these robots are required to traverse following a Hamiltonian path to reduce energy consumption and time requirements. This problem can be formulated as a Travelling Salesman Problem (TSP), an NP-hard problem. Moreover, upon urgent requirements, these robots must traverse in real-time, demanding speedy path planning from the TSP instance. Among the well-known optimization techniques for solving the TSP problem, Ant Colony Optimization has a good stronghold in providing good approximate solutions. Moreover, ACO not only provides near-optimal solutions for TSP instances but can also output optimal or near-optimal solutions for many other demanding hard optimization problems. However, most of the implementations of Ant Colony Optimization on quantum or hybrid quantum architecture proposed in the literature require conversion of classical data to qubits before being fed to the algorithm, and cannot be automated. But quantum-enabled mobile robots require automated path formation after receiving the commands from the environment. The novelties of the proposed work are many-fold. Firstly, the proposed work allows ACO to be applied as its classical counterparts, allowing automation in path formation in quantum-enabled mobile robots. Secondly, this allows a new way of incorporating quantum processing unit in the research of quantum-enabled mobile robots. Researchers around the globe have been trying to incorporate quantum computing in autonomous mobile robots, and true to the best of authors' knowledge, no work in path planning for multiple targets in quantum-enabled mobile robots have been found in literature. Thirdly, quantum processing unit has been applied at exactly that point where it will be most useful, as in NISQ era quantum computer is not reliable for arithmetic processing. Simulation results of the proposed Hybrid Quantum Ant Colony Optimization algorithm on several TSP instances have shown promising results with average error percentage from optimum

results of only 6.985%. Hence, it is expected that the proposed work to be important in future research of fusing the two rising domains of quantum computing and autonomous mobile robots.

Index Terms—Mobile IoT devices, path planning, travelling salesman problem, ant colony optimization, quantum computing.

I. INTRODUCTION

AS WE are entering the future of automation, more and more robotic devices are being used in logistics industry. Firstly, applications of robotic devices allow for reduced error. Secondly, applying robotic automation allows faster calculations of optimized solutions when necessary. These facts result in increased productivity. In fact, robotic delivery services have already been launched in certain nations [1], [2]. Autonomous robots have also been found in other tasks such as remote monitoring for home environment [3], picking up garbage [4], as vacuum cleaners with a novel mobile sink area allocation strategy [5]. Moreover, these autonomous mobile robots are being used for daily services such as fetching some items or cleaning, to name a few, in households and industries. These devices traverse a set of target locations to provide the necessary service(s). Researchers around the globe have started to find better path-planning policies in several such mobile consumer devices, such as for cleaning robots [6].

For such mobile robots, to provide the services at target locations in real time requires speedy planning of possible paths. Moreover, the robots need to traverse as little distance as possible while visiting each target location. This problem can be formulated as a Travelling Salesman Problem (TSP), a well-known NP-hard problem. The intractability of the Travelling Salesman Problem involves the applications of various optimization techniques that can provide near-optimal routes in a graph of target nodes. The most notable techniques among them include Ant Colony Optimization (ACO). Colorni et al. proposed the possibility of optimization following the natural phenomena occurring in ant colonies in 1991 [7]. Dorigo and Gambardella successfully utilized this concept in optimizing the famous Travelling Salesman Problem in 1997 [8]. Since then, ACO has been utilized with great success in solving multitudes of problems that range from basic optimization up to as diverse as analog design [9], reversible circuit synthesis [10], etc. ACO has found its application quite successfully in consumer home automation

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Mayukh Sarkar and Jitesh Pradhan are with the Department of Computer Science and Engineering, National Institute of Technology Jamshedpur, Jamshedpur 831014, India (e-mail: mayukhsarkar.in@gmail.com; jitpradhan02@gmail.com).

Anil Kumar Singh is with the Department of Computer Science and Engineering, Motilal Nehru National Institute of Technology Allahabad, Prayagraj 211004, India (e-mail: ak@mnnit.ac.in).

Hathiram Nenavath is with the Department of Electronics and Communication Engineering, Indian Institute of Technology Bhilai, Bhilai 491002, India (e-mail: hathiram@iitbhilai.ac.in).

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networks as well, where this promising technique has been used for the implementation of clustering algorithm [11].

With the rise of quantum computational paradigm, scientists all around the globe have been trying to incorporate this powerful computing technique in mobile devices, especially in mobile robots. This approach is still in infancy and true to the best of authors' knowledge, not much work has been done. Cornet et al. [12] have given an overview of quantum technologies used in mobile devices. Liu et al. [13] have employed quantum particle swarm optimization to perform logistic regression health assessment in medical Internet of Things (MIoT). Atchade-Adelomou et al. [1] have employed quantum annealing based approach in mobile robots for optimized order picking. In 2022, Chella et al. [14] have employed techniques based on Grover's search algorithm for path planning towards a single goal, given the description of the world with impenetrable obstacles. In 2023, Chella et al. [15] have improved upon the previous approach for a swarm of robots. In 2024, Singamaneni et al. [16] have employed quantum key distribution technique to improve cloud security, that can be utilized in consumer applications. In the same year, Cao et al. [17] have proposed a novel method for software resource allocation of requested services in Digital Twin empowered networks for quantum-enabled consumer applications. Also, in the same year of 2024, Awan et al. [18] have incorporated Ubiquitous AI, Machine Learning, and Quantum Computing in trust-enhanced consumer applications for Software Defined Networks. But true to the best of authors' knowledge, no work has been done in finding optimum path to provide services at multiple target locations, and returning to the base location. Current paper attempts to bridge this gap by solving the Travelling Salesman Problem on a hybrid classical-quantum computer using Ant Colony Optimization, thereby proposing another way to incorporate quantum computational advantages in mobile robots.

Quantum computing started its journey from the proposal of a quantum equivalent of the Turing machine by Benioff [19], [20]. In 1982, Richard Feynman postulated that simulating quantum physics on classical hardware would be impossible and suggested using quantum mechanical hardware for more efficient computation [21]. The real power of quantum computing technique came into view when Peter Shor, in 1994, proposed quantum algorithms for prime factorization and discrete logarithm problems, hence showing the possibility of breaking RSA and Diffie-Hellman encryption techniques [22]. In 1996, Grover showed the possibility of speed-up of search in an unstructured database using quantum system [23]. These results immediately brought quantum computing to the forefront of futuristic computational technique research. The declaration of quantum supremacy by Google using their Sycamore processor [24], and the possibility of universal quantum logic operations with high average fidelity [25] has raised the curtain of those possibilities which have been only theoretical concept for long.

The current paper attempts to propose a novel and automated path-planning technique in quantum-enabled mobile robots, thus progressing another step towards successful incorporation of quantum computing in autonomous mobile robots.

The rest of the paper is organized as follows. Section II provides the necessary overview of quantum computing and the ant colony optimization algorithm. Several attempts have been made to provide a quantum version of ACO in literature. Section III provides a brief overview of previous attempts and the novelties of the current paper. Section IV demonstrates the novelties of the proposed algorithm in details. Section V demonstrates the complete Hybrid Quantum Ant Colony Optimization algorithm as proposed in this paper, step-by-step, with an example of an instance of TSP. Section VI shows the performance of the proposed algorithm on several instances of the TSP obtained from a well-known collection of TSP instances, namely TSPLIB [26], as well as on randomly generated 50 graph instances. The result shows that the proposed algorithm can provide optimal or near-optimal solutions to TSP problems without any manual intervention, as targeted. Section VII finally concludes the paper.

II. BACKGROUND INFORMATION

In this section, required background information on quantum computing and ant colony optimization is being demonstrated.

A. Quantum Computing

In classical computing, the basic unit of information, i.e., a bit, can stay in one of the two states, either 0 or 1, at any particular time. But in a quantum computer, a *qubit* can stay as a *superposition* of the two basis states $|0\rangle$ and $|1\rangle$, where state $|0\rangle$ corresponds to classical state 0 and state $|1\rangle$ corresponds to classical state 1. Mathematically, quantum state of a qubit can be represented as a vector $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \in \mathbb{C}^2$, where α and β are amplitudes associated with the states $|0\rangle$ and $|1\rangle$ respectively. Upon measurement, the quantum state collapses to one of the basis states and outputs the corresponding classical state. As for example, for the above-mentioned quantum state, upon measurement, the quantum state will collapse to state $|0\rangle$ and will output classical bit 0 with probability $|\alpha|^2$, and it will collapse to state $|1\rangle$ and will output classical bit 1 with probability $|\beta|^2$. Thus, the basis state $|0\rangle$ is associated with the vector $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and basis state $|1\rangle$ is associated with the vector $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. An arbitrary quantum state $|\psi\rangle$ can thus be represented as follows.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (1)$$

Moreover, as the measurement of a qubit should always result in either 0 or 1, it must follow that $|\alpha|^2 + |\beta|^2 = 1$.

On a multiqubit system, the qubits involved in the system remain entangled to each other. As an example, on a two-qubit system, the state $|\psi\rangle$ becomes the superposition of all possible two-bit basis states $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$, and can be written as follows.

$$|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle \quad (2)$$

In the above-stated equation, it satisfies $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$. On measurement of both qubits, the

classical result will be either 00, 01, 10 or 11 with the probabilities equal to the square of their corresponding amplitudes. Upon measuring a single qubit, as for example the first qubit, it will output 0 with probability $|\alpha_{00}|^2 + |\alpha_{01}|^2$ and will output 1 with probability $|\alpha_{10}|^2 + |\alpha_{11}|^2$. In case the first qubit measures to 0, the complete quantum system collapses to the state $\frac{\alpha_{00}}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} |00\rangle + \frac{\alpha_{01}}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} |01\rangle$. The similar post-measurement state can be obtained in case the first qubit measures to 1.

The state of a qubit gets modified by quantum gates, and a linear sequence of quantum gates eventually forms a quantum circuit. There cannot be any fanout or loop in a quantum circuit. The state of a quantum system can be manipulated using an adequately designed sequence of gates to harness the inherent parallelism obtained from the superposition property. A quantum gate must be a unitary matrix, and conversely, any unitary matrix is well-suited for a quantum gate. Only a small set of quantum gates is majorly used in quantum architectures, especially the universal set of quantum gates. There are uncountably many possible sets of universal quantum gates. As an example, a Clifford gate set $\{H, S, CNOT\}$, along with a non-Clifford gate such as T gate, is a well-known universal gate set. Now, any single qubit gate U can be represented as a sequence of three rotational gates up to a global phase factor, i.e., for any unitary U , there exist four real numbers α, β, γ , and δ , such that

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta) \quad (3)$$

As all gates in the mentioned gate set are single qubit gates except $CNOT$ gate, it can be said that any functionality can be implemented using two rotation gates and $CNOT$ gate. A complete overview of the basics of quantum computing, along with the proof of the above representation, can be found in [27].

In the proposed work, a quantum circuit will be utilized to produce a quantum state provided the probabilities of nodes to be selected, and then will be measured to obtain the next node. As the probabilities involved in the ACO are real, the circuit will need to generate only those quantum states that are in real vector space. This allows us to restrict ourselves to only R_y and $CNOT$ gates. The quantum selector proposed in [28] has been utilized here for the next node selection and is built up of only controlled- R_y and $CNOT$ gates. An overview of the proposed quantum selector has been given in Section V-A.

B. Ant Colony Optimization

Ant Colony Optimization (ACO) is a well-known evolutionary algorithm that can obtain optimal or near-optimal solutions to many demanding optimization problems. This technique was first proposed in [7]. In this technique, multiple ants move through some path in a graph in order to obtain optimal solutions. Each ant determines its path as follows.

- 1) If there can be multiple possible start nodes, the ant can start from any node randomly, or otherwise, it will start from the root. Then, it follows a path from the starting node as long as it does not reach some solution or as long as it does not reach the leaf node,

depending on the problem statement. While sitting at a node, the ant always tries to find the optimum next node probabilistically. This probability of the subsequent nodes depends on two criteria: firstly, the likeliness of the path, which generally is some function of the path's cost. In a minimization problem, the likeliness varies inversely with the cost; in a maximization problem, the likeliness varies proportionately with the cost. The second selection criterion represents the experience of previous ants in finding paths for optimal or near-optimal solutions. For the first ant, this factor is initialized to the same value for all edges. This factor is commonly known as *pheromone*. Now, let from the node a , the selection of the next node b involves the likeliness η_{ab} and the pheromone τ_{ab} . Then, the probability of the node will be calculated as in equation (4).

$$P(a \rightarrow b) = \frac{\tau_{ab}^\alpha \eta_{ab}^\beta}{\sum_{i \in N(a)} \tau_{ai}^\alpha \eta_{ai}^\beta} \quad (4)$$

α and β are two hyperparameters that needs to be tuned for better results, and sum in the denominator involves all neighbours of node a ($N(a)$ represents the neighbour set of node a).

- 2) Once the ant completes its tour, the edges' pheromones change along with the success of the ant following the rule as in equation (5).

$$\eta_{ab} = (1 - \rho) * \tau_{ab} + f(w) \quad (5)$$

In this equation, ρ is the evaporation rate of the paths, and $f(w)$ is the predetermined function of the overall cost of the final solution found if the ant is successful. If the ant fails to reach any solution, $f(w)$ is taken as zero. If the ant is successful, $f(w)$ varies inversely with w for minimization problems and proportionately for maximization problems. In simpler terms, if the ant fails, the pheromones will only be evaporated, and nothing will be deposited. Furthermore, if the ant succeeds, some amount of pheromone gets deposited depending on the likeliness of the solution found, i.e., the better the solution, the higher the amount of pheromone deposited. For a large number of ants, this eventually allows all ants to converge toward the optimal solution by depositing more pheromones over the optimal paths. This section demonstrates the simple implementation of ACO. The same has been implemented in the current paper for quantum-enabled mobile robots. However, for large instances on a classical computer, faster variations of ACO has been proposed by Ismkan [29].

III. RELATED WORKS

Wang et al. [30], [31] introduced the novel concept of implementing Ant Colony Optimization on a Quantum Computer. In this work, the authors have implemented discrete binary ACO by representing the pheromone, i.e., the probability of the selected path, by qubits. A similar approach of quantum ACO has been employed in implementations of collision detection in virtual environment [32],

privacy-preserving data mining algorithm [33], optimization of evacuation path [34], rule-based fuzzy classifier [35]. Wang et al., in the same paper [30], have also proposed a quantum-inspired ACO employing the same approach, and has been utilized in link prediction [36]. A similar approach to quantum-inspired ACO has been employed in optimizing queries in distributed database systems [37]. The major problem of these works is the updation of the qubit by rotation gates, whose rotation angle is determined by a function of the probability amplitudes of the qubit being applied. This technique, thus, involves knowing the state of the qubit. However, on real quantum hardware, measuring a qubit would collapse its state, and hence, these algorithms cannot be implemented on a real quantum computer.

A different approach to quantum implementation of ACO has been performed by representing the ant positions by coordinates on Bloch sphere [38], [39]. A similar idea has also been employed in various optimization problems, such as in path planning of automated guided vehicles [40]. However, the major problem with this approach is its lack of consistency with the original concept of Ant Colony Optimization.

Ghosh et al. [41] have proposed another novel approach to implementing ACO on a quantum computer named MNDAS. This paper employs a set of qubits to encode all paths, a separate set of qubits to represent pheromones, and three qubits as registers, requiring many qubits in a sufficiently large instance of graphs. Moreover, the proposed architecture in the paper employs a large number of SWAP gates, resulting in the tendency of introducing noise [42]. Thus, such architecture is not implementable in current Noisy Intermediate-Scale Quantum (NISQ) era quantum computers.

de Andoin and Echanobe [42] have proposed an implementable ACO on a hybrid quantum system. The algorithm in this paper also does not follow the conventional approach of ACO and hence faces problems in constrained optimization problems. The paper proposes generating valid solutions from invalid ones in such cases, but that requires the knowledge of the complete set of valid solutions, which is impossible in most optimization problems.

Several works have been done in the implementation of path planning on classical mobile robots [43], [44], [45], [46]. Some works have been done for path-planning in single target location for a quantum-enabled mobile robots have been done as well. [14], [15]. But, true to the best of authors' knowledge, no such work has been done for multiple targets on quantum-enabled mobile robots. In the current paper, the authors have proposed a novel implementation of ACO on a hybrid quantum computer, that can be employed for the same in quantum-enabled mobile robots. After receiving the target locations from the environment, the mobile robot must determine the optimal path without any manual intervention. But as discussed in this section, true to the best of authors' knowledge, all quantum or hybrid quantum implementation of ACO in literature, requires manual creation of qubits from problem instance elements. The novelty of this approach has been demonstrated in Section IV.

IV. NOVELTIES OF THE PROPOSED WORK

The proposed approach of implementing Ant Colony Optimization on a hybrid quantum computer has several novelties demonstrated as follows.

- 1) True to the best of the authors' knowledge, not much work has been done in the domain of incorporating quantum computing in mobile robots. However, an automated technique is necessary to find the optimum or near-optimum path in futuristic quantum-enabled mobile robots, as the mobile robots must automatically perform the optimum path formation after receiving the path information from the environment. But, all the works proposed in quantum-based Ant Colony Optimization require some manual process to be involved, such as representing some part of the instances, such as pheromone or ants, to be represented in quantum bits. This hinders these techniques from being employed in path-finding in quantum-enabled mobile robots. In the proposed work, in the other hand, instance elements do not need to be converted into quantum bits. Instead, once the targets are obtained, a quantum-enabled Ant Colony Optimization is executed with the target goals, where the classical processor performs the arithmetic calculations and sets the parameters of a quantum circuit. When the parameters are set, a zero-initialized quantum circuit gets initiated to calculate the probabilities automatically. At the end of the measurement device, the classical computation resumes again with the measured classical result for the next algorithm phase. This continues as long as a complete cycle covering all nodes is not found. Hence, using the proposed algorithm, a hybrid classical-quantum device can automatically output the final path once the target goals are input.
- 2) Quantum computing, being susceptible to noises from various sources including Earth's magnetic field [47], do not suggest to have qubits travelled far. Hence, quantum computing techniques, at the time of this writing, should be used only when it can really shine. In other words, if quantum computer cannot perform certain task better than the classical counterpart, that task should be delegated to classical processor. One area, applicable in current scenario, where quantum computing will be useful over classical counterparts, is the ability to generate states with pure randomization by a single measurement. A quantum circuit can perform such task inherently due to its superposition properties. Upon measurement, a quantum state collapses to one its bases with pure probabilities based on their amplitudes. Among the steps of Ant Colony Optimization, quantum computing would incorporate noise in the steps involving arithmetic calculations, and hence they have been delegated to a Classical processor. Classical processor will be able to perform arithmetic calculations without adding unnecessary noise in the system. Whereas, QPU has been incorporated at exactly that step where its application would be advantageous over classical processor.

- 3) As has been mentioned in Section I, though some optimization problems such as order-picking [1], path planning in a world with single goal and multiple obstacles [14], [15], have been solved for quantum-enabled mobile devices, but true to the best of authors' knowledge, current paper is the first one to approach optimum and automated path-finding for multiple targets in quantum-enabled mobile robots.

It is expected that the proposed work will be proved useful in future quantum-enabled mobile robots for automatic optimum path finding incorporating the Quantum Processing Unit.

V. PROPOSED IMPLEMENTATION OF ANT COLONY OPTIMIZATION ON HYBRID QUANTUM SYSTEM

This section demonstrates the proposed implementation of the Ant Colony Optimization (ACO) algorithm on a hybrid Quantum-Classical system. Firstly, all the steps involved in the ACO algorithm can be implemented on a quantum computer because any classical algorithm can be simulated on a quantum computer efficiently using Fredkin Gate [27]. Moreover, several implementations of classical logical and arithmetic operations on a quantum computer have been proposed in literature [48], [49]. However, in the current NISQ era, quantum computers are not fault-tolerant enough, and they can lose coherence before executing for sufficiently long [50]. Thus, the algorithms employing a quantum-classical hybrid approach in solving problems rather than employing quantum hardware solely, such as Variational Quantum Eigensolver (VQE) [51] and Quantum Approximation Optimization Algorithm (QAOA) [52], have made immediate impact in this NISQ era of quantum computation.

The current section elaborates on the proposed algorithm as follows. Firstly, Section V-A demonstrates the design of a quantum selector that, upon providing a set of probabilities adding up to 1, will select one of the elements respecting their corresponding probabilities. The proposed HQACO algorithm utilizes this selector to select the next node. The complete step-by-step details of HQACO, with an example of the Travelling Salesman Problem, have been demonstrated in Section V-B.

A. Design of the Quantum Selector

In [28], the author has demonstrated an algorithm that given a set of probabilities $\{p_i\}$ such that $\sum p_i = 1$, will generate a parameterized quantum circuit generating the output state $\sum \sqrt{p_i} |i\rangle$. As an example, given set of probabilities $\{0.3, 0.2, 0.4, 0.1\}$, the algorithm will generate a two-qubit circuit producing the state $(\sqrt{0.3} |00\rangle + \sqrt{0.2} |01\rangle + \sqrt{0.4} |10\rangle + \sqrt{0.1} |11\rangle)$. It can be observed that, upon measurement of the two qubits, the classical output 00 will appear with probability 0.3 , 01 will appear with probability 0.2 , 10 will appear with probability 0.4 and 11 will appear with probability 0.1 . In this subsection, a brief overview of the algorithm is being provided. Detailed discussion with its correctness can be found in [28].

- 1) The spherical coordinates are generated from the set of amplitudes employing Algorithm 1. As for example,

Algorithm 1: GetParameters

input : Array of amplitudes $A = \{a_1, a_2, \dots, a_n\}$ of length n

output: Array of spherical angles $R = \{r_1, r_2, \dots, r_{n-1}\}$ of length $n - 1$

for $i \leftarrow 1$ **to** $n - 1$ **do**

if a_i is 0 **then**

 Append 0 to R .

 Fill the rest of R with π .

break.

end

else

 Set $r_i \leftarrow 2 * \arccos(a_i)$.

 Divide all elements of A by $\sin(r_i/2)$.

end

end

$$|0\rangle \xrightarrow{R_y(r_1)} |\psi\rangle$$

Fig. 1. Single-qubit arbitrary state generator.

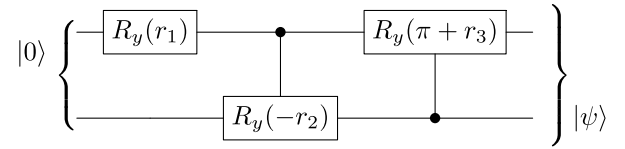


Fig. 2. Two-qubit arbitrary state generator.

from the set of amplitudes $\{\sqrt{0.3}, \sqrt{0.2}, \sqrt{0.4}, \sqrt{0.1}\}$, the algorithm will generate the spherical coordinate angles $\{r_1, r_2, r_3\} = \{1.982313, 2.013707, 0.927295\}$. It can be readily verified that

$$\cos\left(\frac{r_1}{2}\right) = \sqrt{0.3} \quad (6)$$

$$\sin\left(\frac{r_1}{2}\right)\cos\left(\frac{r_2}{2}\right) = \sqrt{0.2} \quad (7)$$

$$\sin\left(\frac{r_1}{2}\right)\sin\left(\frac{r_2}{2}\right)\cos\left(\frac{r_3}{2}\right) = \sqrt{0.4} \quad (8)$$

$$\sin\left(\frac{r_1}{2}\right)\sin\left(\frac{r_2}{2}\right)\sin\left(\frac{r_3}{2}\right) = \sqrt{0.1} \quad (9)$$

- 2) For single-qubit circuit generating 2-dimensional vector, the circuit will be a single R_y gate with the angle obtained in step 1. It can be observed that, from a 2-dimensional amplitude vector $|\psi\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$, step 1 will generate a single angle r_1 such that $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \cos(\frac{r_1}{2}) \\ \sin(\frac{r_1}{2}) \end{bmatrix}$. The circuit is as shown in Figure 1.
- 3) For two-qubit circuit generating 4-dimensional statevector $|\psi\rangle = [a_1 \ a_2 \ a_3 \ a_4]^T$, first the three parameters $\{r_1, r_2, r_3\}$ will be generated following Algorithm 1. Then the circuit generating $|\psi\rangle$ will be as in Figure 2.
- 4) For any q-qubit circuit with $q > 2$ generating 2^q -dimensional statevector $|\psi\rangle$, the $2^q - 1$ dimensional vector of spherical angles will be generated following Algorithm 1. The spherical angle vector will be provided as the $2^q - 1$ parameters to the state generator circuit as

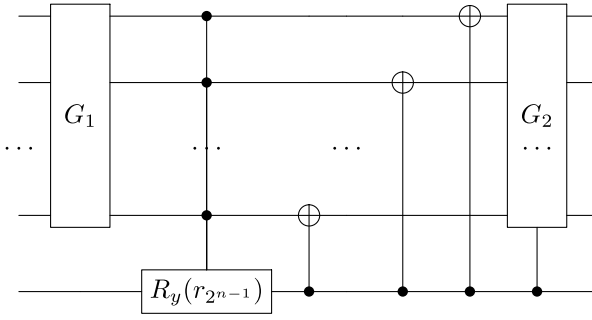


Fig. 3. Multiqubit arbitrary state generator.

shown in Figure 3. In the circuit, G_1 is a recursive $(q-1)$ -qubit arbitrary state generator circuit with first $(2^{q-1} - 1)$ parameters $\{r_1, r_2, \dots, r_{2^{q-1}-1}\}$, the multicontrolled R_y gate has all the above $(q-1)$ lines as controls and employs $(2^{q-1})^{th}$ parameter. Next, there is a group of $(q-1)$ CNOT gates, each having control on q^{th} line and target to each of the above $(q-1)$ lines respectively. Finally, there is another recursive $(q-1)$ -qubit arbitrary state generator circuit G_2 with last $(2^{q-1} - 1)$ parameters $\{r_{2^{q-1}+1}, \dots, r_{2^q-1}\}$. Each gate on G_2 has an additional control from the q^{th} line.

It can be noted that, if the size of the probability vector is not power of 2, corresponding angles generated by Algorithm 1 will be zero, according to *if* block of the algorithm. When these zero values will be set as the parameters to the rotation gates, corresponding gates can be altogether removed or replaced with non-parametric simpler gates, such as when $r = 0$, all $R_y(r)$, $R_y(-r)$ and their controlled counterparts can be omitted altogether, and $R_y(\pi + r)$ gate becomes $R_y(\pi)$ gate.

Employing the stated algorithm, a 3-qubit arbitrary state generator circuit can be designed as in Figure 4, where the parameters are generated from the input 8-dimensional statevector $\{a_1, a_2, \dots, a_8\}$. The parameters $\{r_1, r_2, \dots, r_7\}$ are generated from the statevector by employing the Algorithm 1. The circuit generated using the stated algorithm can generate statevectors with only real amplitudes, which will be sufficient for the proposed HQACO algorithm, as the probabilities generated in the intermediate stages are always real numbers.

The stated circuit requires number of gates exponential to the number of qubit lines, and hence polynomial to the length of the input amplitude vector, for generating exact amplitude [28].

B. HQACO Algorithm

In this subsection, the quantum selector circuit designed in Section V-A will be employed to perform Ant Colony Optimization on a hybrid quantum computer. In demonstrating the algorithm, an instance of the Travelling Salesman Problem on a small graph, as shown in Figure 5, will be employed.

- 1) A parametrized quantum selector with $q = \max(\lceil \log_2(v) \rceil, \lceil \log_2(d) \rceil)$ qubits needs to be prepared with the structure as mentioned in Section V-A, where v is the number of nodes, and d is the maximum degree (outdegree for directed graph) of the graph.

The given example graph has seven vertices, and the maximum degree is 4. Thus, it is sufficient to have a 3-qubit quantum selector, as depicted in Figure 4. If the number of possible candidates, and hence, the number of probabilities to choose from, is less than 2^q (8, in the example), append 0 at the end of the amplitude array to make it of length 2^q .

- 2) The graph information is stored in a classical register C_1 . Each edge has to be initialized with a pheromone value, which will be kept in another classical register C_2 . Initially, all edges receive the same pheromone value. The initial pheromone value has been taken as 0.5 for all the edges in the given example.
- 3) Suitable values of the hyperparameters α , β , and pheromone evaporation rate ρ are chosen and kept in classical register C_4 . In the given example, let us choose the values $\alpha = 0.6$, $\beta = 0.4$ and $\rho = 0.01$ respectively. This will allow the selection to be more directed by pheromone values rather than edge costs.
- 4) Perform the following sequence of operations for each ant.
 - a) The start node for the ant is to be selected with equal probability. The parameters of the quantum selector are set so that the probability of each node becomes $\frac{1}{v}$. In the example with 7 nodes, the 7 parameters of the quantum selector will be the angles obtained using Algorithm 1 from the set of amplitudes $\{\sqrt{\frac{1}{7}}, \sqrt{\frac{1}{7}}, \sqrt{\frac{1}{7}}, \sqrt{\frac{1}{7}}, \sqrt{\frac{1}{7}}, \sqrt{\frac{1}{7}}, \sqrt{\frac{1}{7}}, 0\}$. After calculating the parameters, the classical processor will set them to the quantum selector circuit for further processing.
 - b) Quantum registers are initialized with $|0\rangle$. The quantum selector will produce the desired state. All the qubits of the selector will be measured and stored in a classical register C_3 . The classical register holds the start node index with equal probabilities. Suppose the initial node has been measured as 4 in the example considered.
 - c) The classical processor will read the node from the register C_3 and calculate the probabilities of the next transitions. To calculate this, the classical processor will store all unvisited neighbors and consider the inverse of the corresponding edge costs as the corresponding weights. The probability of a particular incident edge e is then evaluated as $\frac{\tau_e^\alpha \eta_e^\beta}{\sum_{i \in N(a)} \tau_e^\alpha \eta_e^\beta}$, where $\tau(e)$ and $\eta(e)$ are current pheromone values and edge weights respectively, and d is the degree of the current node. As for example, with start node 4 and all pheromone values 0.5, the probability of going to neighbor nodes $\{2, 3, 5, 6\}$ are $\{0.32625, 0.19402, 0.24725, 0.23247\}$, as the corresponding edge costs are $\{\frac{1}{3}, \frac{1}{11}, \frac{1}{6}, \frac{1}{7}\}$ respectively. For current example, the cost has been taken as the inverse of the edge weight, as it is a minimization problem.

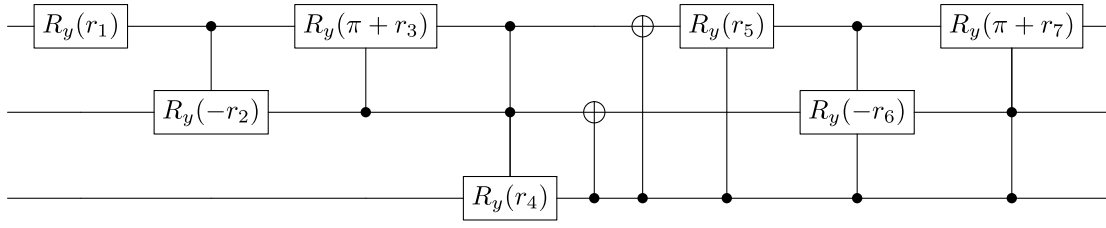


Fig. 4. Example of 3-qubit arbitrary state generator.

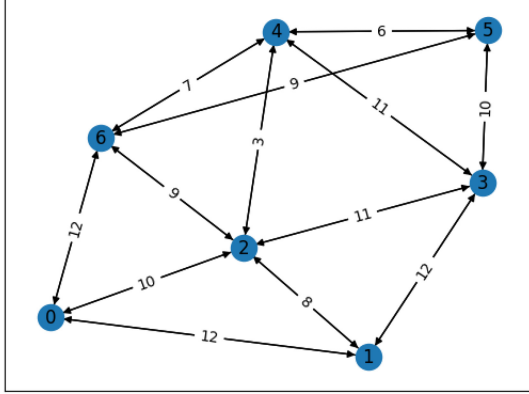


Fig. 5. An Example Graph for HQACO.

- d) The angles are evaluated using Algorithm 1 from the zero-appended amplitude set, where each amplitude is the square root of the corresponding probabilities. Thus, in the continuing example, the classical processor will set the parameters of the quantum selector after evaluating them from the amplitude set $\{0.57118, 0.44048, 0.49724, 0.48215, 0, 0, 0, 0\}$.
- e) The qubits are set in the quantum selector to $|0\rangle$, and are allowed to pass through the quantum selector with set parameters. At the end, the qubits are measured to obtain the next node index. For example, if the measurement result turns out to be 010, the node at index 2 (0-indexed), i.e., node 5, will be selected as the next node. In such scenario, the index 2 is set in classical register C_3 .
- f) Continue the steps 4c to 4e, as long as no unvisited next node can be found. After completion of the tour, the pheromone of each edge is reduced by pheromone evaporation rate ρ following equation (5).
- g) Two events can occur at this point: firstly, the ant has visited all nodes, in which case the ant is said to be successful, or secondly, the ant is stuck without any unvisited neighbor, but all nodes have not been visited. The tour is a failure in the second scenario, and the result will be discarded. In the first scenario, each edge's pheromone on the successful path is increased by $\frac{1}{c}$, where c is the total cost of the path.
- h) If a Hamiltonian cycle is to be obtained, another condition needs to be added for a path to be

successful, and that is, the last node on the path must have the starting node as its neighbor.

In the given example, the above step has been performed with 100 ants, and the minimum Hamiltonian cycle $\{6, 5, 3, 1, 0, 2, 4, 6\}$ has been obtained with cost of 63.

C. Expansion to Parallel Ant Systems

The proposed HQACO algorithm can be expanded naturally to parallel ant systems, where multiple ants will start in parallel rather than sequentially at each iteration. On a classical computer, this is achieved by multithreads, which will be executed on multiprocessors or multicores, whichever is available. On a quantum computer, the same can be achieved by placing multiple quantum co-processors, each co-processor being selected by each ant in parallel. Thus, in such a system, each ant will have a separate unit of classical-quantum hybrid processor. Moreover, such parallel classical-quantum hybrid systems need to have shared memory. If some ant successfully reaches the target, i.e., finds a solution, it can update its solution to the shared memory. Next ants, while searching for optimum path, can prune their search in case the visited edge list already crosses the stored result found by another ant. In this way, ants may communicate among themselves on the parallel hybrid classical-quantum system via shared memory.

VI. EXPERIMENTAL RESULTS

This section demonstrates the experimental results of the proposed HQACO algorithm on several instances of the Travelling Salesman Problem obtained from TSPLIB [26], a well-known collection of TSP instances for experimentations. The experiment has been performed using Python 3.11 with Qiskit 0.43 in a locally installed Linux environment. For larger instances, the Qiskit provider for Amazon Braket has been used and executed in an EC2 instance of Amazon AWS. The Table I demonstrates the experimental results for 5 TSP instances, with the minimum path found and the overall estimation error. In each instance, 1000 ants have been used with $\rho = 0.01$. For α and β , multiple values have been tested and the best result obtained for each instance have been recorded accordingly. Ten shots have been used during each measurement, and the output with the highest appearance count has been selected, as a large number of shots would make the output nearly deterministic by driving the output towards the highest probability.

In the first instance with 17 nodes, for each ant, a 5-qubit circuit is required to find the starting node, whereas a 4-qubit circuit is sufficient to find the path as each node has a degree

TABLE I
EXPERIMENTAL RESULTS

TSP Instance	Number of Nodes	Lower Bound In Specification [26]	Minimum Cycle Found using HQACO		Error in Estimation	Number of Qubits used
			Cost	Path		
gr17.tsp	17	2085	2130	[12, 16, 6, 7, 5, 13, 14, 2, 10, 9, 1, 4, 8, 11, 15, 3, 0, 12]	2.16%	5 for node selection 4 for path selection
burma14.tsp	14	3323	3547	[3, 4, 5, 11, 6, 12, 0, 7, 8, 10, 9, 1, 13, 2, 3]	6.74%	4
gr21.tsp	21	2707	3216	[2, 8, 4, 15, 11, 3, 0, 5, 7, 6, 18, 19, 10, 16, 9, 17, 12, 13, 14, 1, 20, 2]	18.8%	5
bayg29.tsp	29	1610	1922	[23, 12, 15, 6, 22, 7, 27, 0, 5, 11, 8, 2, 28, 25, 4, 20, 1, 19, 9, 3, 14, 17, 13, 21, 16, 18, 24, 10, 26, 23]	19.38%	5
bays29.tsp	29	2020	2599	[6, 24, 18, 15, 14, 10, 13, 21, 16, 17, 3, 9, 19, 4, 8, 11, 5, 27, 0, 23, 7, 22, 25, 28, 2, 1, 20, 12, 26, 6]	28.66%	5

of 16. In the simulation experiment, two separate circuits have been used for the same. The same can be followed and suggested in real quantum hardware rather than using a single 5-qubit circuit. Though it will increase the build cost, using a 5-qubit circuit, even where a 4-qubit circuit is sufficient, will increase the execution time and probability of noise in output.

The results have been sorted based on errors in estimation, and it can be observed that the error percentage increases with the number of qubits in the circuit. This output is reasonably expected, as the larger instances require more ants to converge, and the first 1000 ants in each case have shown apparent convergence towards a lower value. In the simulation, though larger circuits require exponential time to execute, they are expected to run extremely fast on real quantum hardware, as quantum simulation on classical hardware is exponential. Hence, on real quantum hardware, a much larger number of ants can be utilized to obtain better results.

Along with the well-known TSP instances obtained from TSPLIB [26], the proposed HQACO algorithm has been applied on 50 randomly generated 16-node graphs. As these randomly generated graphs do not have any specification, firstly their optimum solution have been found using Google OR-Tools [53]. Then the proposed algorithm have been employed on the 50 random graphs. During application, the comparison with the OR-Tools result have been made with varying hyperparameters alpha and beta. For each hyperparameter choice, overall average error in estimation from the OR-Tools output have been calculated and reported as in Table II. As can be observed, giving higher weightage towards the edge weights provides better results in this scenario. In fact, in two instances with α and β set with 0.2 and 0.8 respectively has given exact optimum value, and in multiple instances this parameter settings have produced less than 1% error. However, if the value of β is further increased, more weightage towards edge weights will make the algorithm more deterministic, and at each edge, the behavior will be almost equivalent to applying greedy approach to find next neighbor. The number of ants in the following experiments have been constrained to 500. Though increasing the number of ants would lead to better convergence, but keeping it at 500 allows to see the convergence rates of the different α and β parameter settings. It can be observed that, according to expectations, higher weightage towards pheromone values is leading ants to search more randomly, and hence slower

TABLE II
AVERAGE ERROR RECEIVED ON VARIOUS
VALUES OF HYPERPARAMETERS

Various value of α and β	$\alpha = 0.8$	$\alpha = 0.5$	$\alpha = 0.2$
	$\beta = 0.2$	$\beta = 0.5$	$\beta = 0.8$
Average Error Obtained	65.9%	23.534%	6.985%

convergence rate, whereas higher weightage towards edge cost is allowing ants to converge much faster. Though the hyperparameters have been set via experimentation, systematic approach to hyperparameter setting for more sophisticated requirements can be found at [54], [55].

The chosen TSPLIB instances of Table I have further been tested with the optimum values of $\alpha = 0.2$ and $\beta = 0.8$, and have been shown in Table III. And it can be observed that, for some of the chosen five instances, the specific hyperparameter values have not given optimum results. The complete algorithm being stochastic, this can be expected, and hence such experiments are required to perform with various values of hyperparameters, and the optimum one must be chosen among them, as has been done in Table I.

The discussion in this section shows that, the proposed HQACO algorithm can provide desired optimum or near-optimum path quite accurately, thus providing an important pathway of incorporating quantum computing in mobile computing.

VII. CONCLUSION

Current paper proposes a novel implementation of the Ant Colony Optimization algorithm on a hybrid quantum computer. Though there has been several works on the implementation of ACO using quantum computational paradigm, each of them requires manual intervention for formation of qubits from classical problem instance elements. This hinders the use of such algorithms in automated environments, such as path planning in mobile robots. The mobile robots take the target locations from the environment, and it needs to plan the optimum path given the distances between the targets. Though there are several algorithms to solve this problem in classical computation, true to the best of authors' knowledge, no work has been done to perform the same for multiple targets on quantum-enabled mobile robots. Existing quantum solutions of TSP cannot be used for their manual intervention.

TABLE III
EXPERIMENTAL RESULTS WITH $\alpha = 0.2$ AND $\beta = 0.8$

TSP Instance	Number of Nodes	Lower Bound In Specification [26]	Minimum Cycle Found using HQACO		Error in Estimation	Number of Qubits used
			Cost	Path		
gr17.tsp	17	2085	2223	[14, 13, 2, 5, 7, 16, 12, 6, 0, 3, 15, 11, 8, 4, 1, 9, 10, 14]	6.62%	5 for node selection 4 for path selection
burma14.tsp	14	3323	3547	[3, 4, 5, 11, 6, 12, 0, 7, 8, 10, 9, 1, 13, 2, 3]	6.74%	4
gr21.tsp	21	2707	3216	[2, 8, 4, 15, 11, 3, 0, 5, 7, 6, 18, 19, 10, 16, 9, 17, 12, 13, 14, 1, 20, 2]	18.8%	5
bayg29.tsp	29	1610	2277	[18, 14, 3, 15, 26, 23, 27, 0, 20, 1, 8, 11, 2, 19, 28, 25, 4, 5, 7, 12, 9, 17, 21, 16, 13, 10, 24, 6, 22, 18]	41.43%	5
bays29.tsp	29	2020	2846	[0, 23, 15, 18, 12, 9, 19, 8, 27, 4, 11, 5, 25, 28, 1, 20, 3, 14, 13, 17, 16, 21, 10, 24, 6, 22, 26, 7, 2, 0]	40.89%	5

Current work alleviates this problem by incorporating quantum phenomena in that part of the algorithm, where it would really shine, i.e., in the selection of next node, where probabilistic selection is to be made. The reason behind this decision is that, the quantum computer in the NISQ era is not well-known enough to perform speedier arithmetic calculation reliably. Hence, current work is expected to be important in the incorporation of quantum computing techniques in mobile robots.

True to the best of authors' knowledge, this is the first work to perform the same for multiple targets on quantum-enabled mobile robots. In fact, to obtain an output probabilistically, it is not always necessary to generate the quantum state with exact probability amplitudes, as has been done in this paper. Thus, further research can be performed to check if the same can be performed within a tolerable error rate, but with less costlier quantum circuit. Moreover, there are several faster variations of ACO implemented in literature of classical computing [29]. Further research can be carried out to find whether such faster variations can be implemented in similar manner on a quantum computer.

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